

Disproof of Conjecture 00000007964: binary linear quasi-perfect codes exist for every length $n \geq 2$

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Abstract

Conjecture 00000007964 claims, among other things, that binary linear quasi-perfect codes of length n exist if and only if $n = 2^k - 1$ or n lies in a finite exceptional list. This is false. A code is quasi-perfect when its covering radius is one more than its packing radius $e = \lfloor (d-1)/2 \rfloor$. We prove that such codes exist for every length $n \geq 2$ and for no length $n \leq 1$. The witnesses are not degenerate: for every $n \geq 3$ with $n+1$ not a power of two, the shortened Hamming code of length n is a quasi-perfect $[n, n - \lceil \log_2(n+1) \rceil, 3]$ code. For every $m \geq 2$, the extended Hamming code of length 2^m is quasi-perfect with $d = 4$ and has a perfectly uniform shell distribution. The set of lengths outside $\{2^k - 1\}$ is infinite, so no finite exceptional list can rescue the clause. All three code families, the full length spectrum and the refutation of the clause (also restricted to $d \geq 3$ or $d \geq 4$, and in the non-strict reading) are formalized in Lean 4 (core library only).

1 The conjecture and the clause we refute

The English statement reads:

Definition: The classification extension of perfect codes: nontrivial q -ary perfect codes (covering radius 1, balls fill the space) have known types the Hamming codes and the binary/ternary Golay codes; define relaxed quasi-perfect codes (Q-uniform ball measures) with tight uniform shell distribution at covering radius $r + 1$. *Conjecture:* The gap constant of the radius spectrum of quasi-perfect codes is at most 2 (covering efficiency at least $1/2$); and binary linear quasi-perfect codes of length n exist if and only if n is of type $2^k - 1$ or in a finite exceptional list (the Levenshtein set), completely enumerating the spectrum.

The Chinese version says the same thing. The conjecture is a conjunction. Its first conjunct (“gap constant of the radius spectrum”) is never defined, and we do not discuss it. Refuting the second conjunct suffices:

$$\begin{aligned} & \exists \text{ finite } L \subset \mathbb{N} \quad \forall n : \\ & (\exists \text{ binary linear quasi-perfect code of length } n) \iff (\exists k : n = 2^k - 1) \vee n \in L. \end{aligned} \tag{1}$$

Definition 1. A *binary linear code* of length n is an $\mathbb{F}_2$ -subspace $C \subseteq \mathbb{F}_2^n$ with $|C| \geq 2$. Its *minimum distance* is $d = \min\{d(x, y) : x \neq y \in C\}$ (Hamming distance), its *packing radius* is $e = \lfloor (d-1)/2 \rfloor$, and its *covering radius* is $\rho = \max_{x \in \mathbb{F}_2^n} \min_{c \in C} d(x, c)$. The balls of radius e around codewords are disjoint and those of radius ρ cover $\mathbb{F}_2^n$. Hence $\rho \geq e$, and C is *perfect* if $\rho = e$. C is *quasi-perfect* if $\rho = e + 1$.

This is the standard definition (for example, MacWilliams and Sloane). The conjecture writes it as “covering radius $r+1$ ” for packing radius r . Section 7 covers the other readings: the non-strict version $\rho \leq e+1$, a “nontriviality” demand $d \geq 3$ or $d \geq 4$, restrictions on k , and the “relaxed, uniform shell” variant from the conjecture’s preamble. The counterexamples defeat all of them.

2 Syndromes

We use the bitwise exclusive or $a \oplus b$ of nonnegative integers. It is addition in the $\mathbb{F}_2$ -vector space of finitely supported bit strings. In particular it is associative and commutative, $a \oplus a = 0$, $a \oplus 0 = a$, and $a \oplus b = 0 \iff a = b$.

Fix labels $\ell_1, \dots, \ell_n \in \mathbb{N}$, one per coordinate. The *syndrome* of $x \in \mathbb{F}_2^n$ is

$$\sigma(x) = \bigoplus_{j: x_j=1} \ell_j.$$

Then $\sigma(x+y) = \sigma(x) \oplus \sigma(y)$, so $\ker \sigma$ is a linear code. This is just the code with parity-check matrix whose j -th column is the binary expansion of ℓ_j . Write e_j for the j -th unit vector.

Lemma 2. *Suppose the labels are pairwise distinct.*

1. *If $\text{wt}(x) = 0$ then $\sigma(x) = 0$. If $\text{wt}(x) = 1$, say $x = e_j$, then $\sigma(x) = \ell_j$. If $\text{wt}(x) = 2$ then $\sigma(x) \neq 0$.*
2. *If all $\ell_j < 2^m$ then $\sigma(x) < 2^m$ for every x .*
3. *For $x \in \mathbb{F}_2^n$ and y with $\sigma(y) = \sigma(x)$, the word $c = x + y$ lies in $\ker \sigma$ and $d(x, c) = \text{wt}(y)$.*

Proof. (1) $\sigma(e_i + e_j) = \ell_i \oplus \ell_j \neq 0$ for $\ell_i \neq \ell_j$. (2) XOR does not create bits above position $m-1$. (3) $\sigma(x+y) = \sigma(x) \oplus \sigma(y) = 0$ and $x+c=y$. $\square$

Lemma 3. *If $t = 2^k \leq s < 2^{k+1}$ then $s \oplus t = s - t < t$.*

Proof. Bit k is the leading bit of s . XOR with 2^k clears it and leaves the lower bits unchanged. $\square$

3 Shortened Hamming codes: $d = 3$, every length $n \geq 3$ with $n+1 \neq 2^k$

For $n \geq 1$ let $\text{SH}(n) = \ker \sigma \subseteq \mathbb{F}_2^n$ with labels $\ell_j = j$ ($j = 1, \dots, n$). For $n = 2^m - 1$ this is the Hamming code. For other n it is a shortened Hamming code.

Theorem 4. *Let $n \geq 3$ and suppose $n+1$ is not a power of 2. Then $\text{SH}(n)$ is a binary linear code with minimum distance $d = 3$ (so $e = 1$) and covering radius $\rho = 2$. It is quasi-perfect.*

Proof. *Minimum distance.* Let $c \neq 0$ be a codeword. $\text{wt}(c) \neq 1$, because $\sigma(e_j) = j \neq 0$. $\text{wt}(c) \neq 2$ by Lemma 2(1). So $\text{wt}(c) \geq 3$. Since $1 \oplus 2 \oplus 3 = 0$, $e_1 + e_2 + e_3$ is a codeword of weight 3. For a linear code the minimum distance is the least nonzero weight, so $d = 3$.

Covering radius ≤ 2 . Let $k = \lfloor \log_2 n \rfloor$ and $t = 2^k$, so $t \leq n < 2t$. Let $x \in \mathbb{F}_2^n$ and $s = \sigma(x)$. By Lemma 2(2) with $m = k+1$ we have $s < 2t$. We find y with $\text{wt}(y) \leq 2$ and $\sigma(y) = s$:

- if $s = 0$, take $y = 0$;

- if $1 \leq s \leq n$, take $y = e_s$;
- if $n < s < 2t$, put $b = s \oplus t$. By Lemma 3, $b < t \leq n$. Also $b \neq 0$, since otherwise $s = t \leq n$. So $y = e_t + e_b$ is a word of weight 2 with $\sigma(y) = t \oplus b = s$.

By Lemma 2(3), $c = x + y$ is a codeword with $d(x, c) \leq 2$.

Covering radius ≥ 2 . Since $n < 2t$ and $n + 1 \neq 2t$, we have $n \leq 2t - 2$. Put $s^* = 2t - 1$, so $n < s^* < 2t$. By the previous paragraph there is y^* with $\sigma(y^*) = s^*$. For every codeword c , $\sigma(y^* + c) = s^*$. Since $s^* \neq 0$ and $s^* \notin \{1, \dots, n\}$, Lemma 2(1) gives $\text{wt}(y^* + c) \notin \{0, 1\}$. So $d(y^*, c) \geq 2$ for all c , and $\rho = 2 = e + 1$. $\square$

For $n = 2^m - 1$ the same argument gives $\rho = 1 = e$: every $s < 2^m$ is a label or 0. So $\text{SH}(2^m - 1)$ is the perfect Hamming code and not quasi-perfect. The Lean file proves this as a sanity check (`hamming_not_QP`). The dimension of $\text{SH}(n)$ is $n - \lceil \log_2(n+1) \rceil$, because the labels $1, 2, 4, \dots, 2^k \leq n$ make σ surjective onto $\{0, \dots, 2^{k+1} - 1\}$. This is not needed for the disproof.

4 Extended Hamming codes: $d = 4$, lengths 2^m

For $m \geq 2$ let $N = 2^m$. Label the coordinates $0, 1, \dots, N - 1$ and let

$$\text{EH}(m) = \{x \in \mathbb{F}_2^N : \sigma(x) = 0, \text{ wt}(x) \text{ even}\}.$$

This is the extended Hamming code. It is linear because σ and the parity $\text{wt}(x) \bmod 2$ are both linear.

Theorem 5. *$\text{EH}(m)$ has $d = 4$ (so $e = 1$) and covering radius 2, so it is quasi-perfect. Moreover, every word at distance 2 from the code has exactly $N/2$ codewords at distance 2. Every word at distance 1 has exactly one codeword at distance 1 and none at distance 2.*

Proof. A nonzero codeword has even weight, and weight 2 is excluded by Lemma 2(1). Since $0 \oplus 1 \oplus 2 \oplus 3 = 0$, $e_0 + e_1 + e_2 + e_3$ is a codeword of weight 4. So $d = 4$.

Let $x \in \mathbb{F}_2^N$ and $s = \sigma(x) < N$. If $\text{wt}(x)$ is odd, $y = e_s$ has syndrome s and odd weight. If $\text{wt}(x)$ is even and $s \neq 0$, $y = e_0 + e_s$ has syndrome s and even weight. If $\text{wt}(x)$ is even and $s = 0$, x is a codeword. In each case $c = x + y$ is a codeword with $d(x, c) \leq 2$. For $x = e_0 + e_1$ and any codeword c , $x + c$ has syndrome $1 \neq 0$ and even weight, so $d(x, c) \geq 2$. Hence $\rho = 2$.

Shells: if $d(x, C) = 2$ then $\text{wt}(x)$ is even and $s = \sigma(x) \neq 0$. The codewords at distance 2 are the $x + e_a + e_b$ with $a \oplus b = s$, and there are exactly $N/2$ unordered pairs $\{a, a \oplus s\}$. If $d(x, C) = 1$, the codeword at distance 1 is unique, because $d = 4 > 2$. By parity no codeword is at even distance from x . $\square$

The shell numbers depend only on the distance to the code. So $\text{EH}(m)$ is uniformly packed: its shell distribution at the covering radius $e + 1$ is as uniform as it can be.

5 Even-weight and repetition codes

Proposition 6. *For $n \geq 2$ the even-weight code $\text{EW}(n) = \{x : \text{wt}(x) \text{ even}\}$ has $d = 2$, $e = 0$ and $\rho = 1$, so it is quasi-perfect. Every word not in the code has exactly n codewords at distance 1.*

Proof. Nonzero codewords have weight ≥ 2 , and $e_1 + e_2$ is a codeword. A word of odd weight becomes a codeword by flipping any one of its n bits, and it is at odd distance from every codeword. So $\rho = 1$, attained by e_1 . $\square$

For even $n = 2m$ the repetition code $\{0, 1\}$ has $d = 2m$, $e = m - 1$ and $\rho = m$. Every word is within $\min(\text{wt } x, n - \text{wt } x) \leq m$ of a codeword, and weight- m words are at distance m from both codewords. So it is quasi-perfect as well. This family is mentioned only for completeness and is checked in `verify.py`.

Proposition 7. *There is no binary linear quasi-perfect code of length $n \leq 1$.*

Proof. For $n = 0$ there is only one word, so no code has two codewords. For $n = 1$ the only linear code with two codewords is $\mathbb{F}_2^1$. It has $\rho = 0 < e + 1$. $\square$

Theorem 8 (length spectrum). *A binary linear quasi-perfect code of length n exists if and only if $n \geq 2$. Codes with $d = 3$ exist for every $n \geq 3$ with $n + 1 \neq 2^k$ (Theorem 4). Codes with $d = 4$ exist for every $n = 2^m$, $m \geq 2$ (Theorem 5).*

Proof. Use Propositions 6 and 7. $\square$

6 The disproof

Corollary 9. *Clause (1) is false. In fact, already its “only if” direction fails for every finite L . This holds even when only codes with $d \geq 3$ (or $d \geq 4$) are counted, and even when perfect codes are also allowed (reading $\rho \leq e + 1$).*

Proof. Let L be finite and $M = \max(L \cup \{0\})$. Put $n = 2(M + 2)$. Then $n \geq 4$, $n \notin L$, and $n + 1$ is odd and ≥ 5 , so it is not a power of 2. By Theorem 4, $\text{SH}(n)$ is quasi-perfect with $d = 3$, but n is neither of the form $2^k - 1$ nor in L . For the $d \geq 4$ version take $n = 2^{M+2} > M$. Then $n + 1$ is odd and > 1 , and $\text{EH}(M + 2)$ is quasi-perfect with $d = 4$. A code with $\rho = e + 1$ in particular satisfies $\rho \leq e + 1$, so the non-strict reading is covered too. $\square$

So the true spectrum is $\{n \geq 2\}$, and its complement $\{0, 1\}$ is finite. The proposed “spectrum” $\{2^k - 1\} \cup L$ has density zero. The clause seems to confuse quasi-perfect codes with perfect codes. The lengths of nontrivial binary perfect codes do lie in $\{2^k - 1\} \cup \{23\}$, and the binary Golay code supplies the exceptional length.

7 Robustness of the interpretation

- **Strict versus non-strict.** Our codes have $\rho = e + 1$ exactly, so they are quasi-perfect whether or not perfect codes are counted (Lean: `onlyIfWeak_false`).
- **Trivial codes.** Someone could object to $d \leq 2$ or to repetition codes. The shortened Hamming codes ($d = 3$, every admissible $n \geq 4$) and the extended Hamming codes ($d = 4$) are standard nontrivial codes (Lean: `onlyIf3_false`, `onlyIf4_false`, `levenshteinAtLeast_false` for every bound $D \leq 4$ on d).
- **Range of k , and the “if” direction.** We refute the “only if” direction with k arbitrary. This is the weakest form, and it implies every variant that restricts k . The “if” direction is not used. (It holds for $k \geq 2$ because of $\text{EW}(2^k - 1)$, and fails only at $n \in \{0, 1\}$.)
- **The exceptional list.** L is arbitrary and finite, which includes whatever the “Levenshtein set” is meant to be. The counterexample lengths form an infinite set (all even $n \geq 4$ already suffice).

- **The “relaxed quasi-perfect (uniform shell)” notion.** The preamble’s notion is not precisely defined. Read it as “ $\rho = e + 1$ and the number of codewords in each shell around a word depends only on that word’s distance to the code”. Then $\text{EH}(m)$ (Theorem 5: shell counts 1, 0 and $N/2$) and $\text{EW}(n)$ (Proposition 6: shell count n) qualify. Their lengths, 2^m and all $n \geq 2$, again lie outside $\{2^k - 1\} \cup L$ for infinitely many n . Shortened Hamming codes are in general *not* uniformly packed, so under this reading the disproof rests on EH and EW. The shell counts are proved above and checked by `verify.py`. They are not formalized in Lean.
- **Linear.** Our codes are $\mathbb{F}_2$ -subspaces. Linearity is part of the Lean definition.

8 Lean formalization

The project `lean4/` (Lean 4.19.0, core library only) defines words as `List Bool`, with weight `wt`, coordinatewise sum `xorW`, distance `dist` and the syndrome `synd i x`, the XOR of the labels $i, i + 1, \dots$ of the 1-positions. A code is a predicate `C : List Bool -> Prop`:

- `IsLinear n C`: all codewords have length n , $0 \in C$, and C is closed under `xorW`.
- `MinDist C d`: all distinct codewords are at distance $\geq d$, and some pair is at distance exactly d .
- `CovRad n C r`: every word of length n is within r of C , and some word is at distance $\geq r$ from all of C .
- `IsQP n C`: `IsLinear n C` and, for some d , `MinDist C d` and `CovRad n C ((d-1)/2+1)`.

Lemmas `minDist_unique` and `covRad_unique` show these are genuine (unique) values. The main theorems are:

- `SH_quasiPerfect`: Theorem 4 for all n . It uses `xor_top_lt` (Lemma 3, proved via `Nat.testBit`).
- `EH_quasiPerfect`: the quasi-perfect part of Theorem 5 for all $m \geq 2$.
- `EW_quasiPerfect`: Proposition 6 without the shell count. `no_QP_short`: Proposition 7.
- `hamming_not_QP`: the Hamming codes of length $2^m - 1$, $m \geq 3$, are not quasi-perfect. This shows that `IsQP` is not vacuous in the other direction.
- `QP_spectrum`: `QPLen n  $\iff$  2 <= n`.
- `conjecture_00000007964_false`: $\neg \text{LevenshteinClause}$, where `LevenshteinClause` is $\exists L : \text{List Nat}, \forall n, \text{QPLen } n \iff ((\exists k, n + 1 = 2^k) \vee n \in L)$.
- `onlyIf3_false`, `onlyIf4_false`, `onlyIfWeak_false` and `levenshteinAtLeast_false`: the variants of Corollary 9.

The project contains no `sorry`, `native_decide` or added axioms. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. Not in Lean: the shell counts of Theorem 5 and Proposition 6, the repetition codes, and the dimension formula.

9 Independent computer check

`verify.py` (Python standard library) works by brute force. It lists codewords explicitly, computes d as the least nonzero weight, and computes ρ by a multi-source breadth-first search on the hypercube. It confirms:

- $\text{SH}(n)$ for $3 \leq n \leq 16$: $d = 3$, with $\rho = 2$ except at $n \in \{3, 7, 15\}$, where $\rho = 1$ (Hamming codes).
- $\text{EH}(m)$ for $m = 2, 3, 4$: $d = 4$, $\rho = 2$, and the shell counts of Theorem 5.
- $\text{EW}(n)$ for $2 \leq n \leq 14$, and the even repetition codes up to length 16.
- An exhaustive enumeration of all binary linear codes of length $n \leq 7$. There are 1, 2, 5, 16, 67, 374, 2825, 29212 of them, including $\{0\}$. Quasi-perfect codes exist exactly for $n \geq 2$ in this range. Those with $d \geq 3$ exist for $n = 4, 5, 6, 7$ (5, 15, 107 and 1260 codes).
- Lemma 3 for $k \leq 12$.

10 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, no Mathlib
cd .. && python3 verify.py  # about one minute
pdflatex report.tex && pdflatex report.tex
```